

The Mixing Time of the Top-to-Random Shuffle

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1 Introduction

This paper is the final report for my Directed Reading Program, in which I worked through portions of *Markov Chains and Mixing Times* by Levin and Peres [1] with my mentor Jason Prodrromidis. Of the many natural questions one can ask about a Markov chain, perhaps the most basic concerns its long-run behavior: for a sufficiently “nice” chain, the distribution at time t converges as $t \rightarrow \infty$ to a limit that does not depend on the starting state. This limit is called the *stationary distribution*. The natural follow-up is to ask about the rate of convergence: how long must the chain run before its distribution is close to stationary? For shuffling cards, this becomes the familiar question of how many shuffles sufficiently randomize a deck, where the answer depends both on the mechanism and on what “random” is taken to mean. The present paper, drawn entirely from [1, Chapters 4, 6, and 7], develops this question for one of the simplest shuffles imaginable: the *top-to-random shuffle*, in which the top card is removed and reinserted at a uniformly random position. We prove sharp upper and lower bounds, in total variation distance, on the number of shuffles required to reach uniformity, and together these bounds establish the *cutoff phenomenon*—a sharply defined threshold near $t = n \log n$ before which the deck is far from random and after which it is essentially uniform.

1.1 Markov chains and the question of mixing

Throughout, $\mathcal{X}$ denotes a finite state space and P a transition matrix on $\mathcal{X}$, that is, a $\mathcal{X} \times \mathcal{X}$ matrix with nonnegative entries whose rows sum to 1. Recall that a sequence of $\mathcal{X}$ -valued random variables $(X_t)_{t \geq 0}$ is a *Markov chain* with transition matrix P if

$$\mathbb{P}\{X_{t+1} = y \mid X_0, \dots, X_t\} = P(X_t, y).$$

A probability distribution π on $\mathcal{X}$ is *stationary* for P if $\pi P = \pi$. Throughout this paper P will be irreducible and aperiodic, in which case π exists and is unique, and the Convergence Theorem [1, Theorem 4.9] asserts the existence of constants $C > 0$ and $\alpha \in (0, 1)$ for which

$$\max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq C\alpha^t.$$

This guarantees convergence to π , but says nothing about the *rate*. Our goal is to quantify it precisely for one specific chain. To do this we first need a notion of distance between probability distributions.

1.2 Total variation distance

Definition 1.1 ([1, Eq. (4.1)]). The *total variation distance* between two probability distributions μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\text{TV}} := \max_{A \subseteq \mathcal{X}} |\mu(A) - \nu(A)|. \quad (1)$$

The right-hand side of (1) is the largest possible disagreement in probability between μ and ν on any single event. While this definition is conceptually clean, it is rarely the most convenient form to work with: one would like to avoid optimizing over all $2^{|\mathcal{X}|}$ events. The following equivalent characterizations, each useful for a different purpose, are central to the rest of the paper.

Proposition 1.2 ([1, Propositions 4.2, 4.5, 4.7 and Eq. (4.5)]). *Let μ and ν be probability distributions on $\mathcal{X}$. Then*

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu(x) - \nu(x)|, \quad (2)$$

$$= \sum_{x: \mu(x) \geq \nu(x)} [\mu(x) - \nu(x)], \quad (3)$$

$$= \frac{1}{2} \sup \left\{ \sum_x f(x) \mu(x) - \sum_x f(x) \nu(x) : \max_{x \in \mathcal{X}} |f(x)| \leq 1 \right\}, \quad (4)$$

$$= \inf \{ \mathbb{P}\{X \neq Y\} : (X, Y) \text{ is a coupling of } \mu \text{ and } \nu \}. \quad (5)$$

Moreover, the infimum in (5) is attained.

A *coupling* of μ and ν is any pair (X, Y) of random variables on a common probability space with marginals μ and ν respectively. Identity (2) reduces a maximum over $2^{|\mathcal{X}|}$ events to a single sum and is the form most amenable to direct computation; (3) restricts that sum to the set where μ dominates and will be convenient in the analysis of separation distance below; (4) interprets total variation distance as the operator norm when distributions act on bounded test functions; and the coupling characterization (5) provides a probabilistic tool for *upper-bounding* the distance, since exhibiting any coupling that often agrees gives an immediate bound. The key idea behind (2) is to take $A = \{x : \mu(x) \geq \nu(x)\}$ as the event achieving the maximum in (1), and the key idea behind (5) is to construct an optimal coupling that “sticks” on the overlap region $\min(\mu, \nu)$ of the two distributions; full proofs are in [1, §4.1–4.2].

1.3 Mixing time

With total variation distance fixed, we measure how far the chain is from equilibrium at time t by the worst-case distance over starting states:

$$d(t) := \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|_{\text{TV}}. \quad (6)$$

Definition 1.3 ([1, Eqs. (4.30)–(4.31)]). The *mixing time* of the chain (with tolerance $\varepsilon \in (0, 1)$) is

$$t_{\text{mix}}(\varepsilon) := \min\{t \geq 0 : d(t) \leq \varepsilon\}, \quad t_{\text{mix}} := t_{\text{mix}}(1/4).$$

The choice $\varepsilon = 1/4$ is conventional: any value strictly below $1/2$ would do, since a standard submultiplicativity argument [1, Eq. (4.34)] gives $t_{\text{mix}}(\varepsilon) \leq \lceil \log_2(\varepsilon^{-1}) \rceil t_{\text{mix}}$, so that knowing t_{mix} controls $t_{\text{mix}}(\varepsilon)$ for every ε up to logarithmic factors.

1.4 The cutoff phenomenon

For many natural Markov chains, $d(t)$ does not decay smoothly: it stays close to 1 until time t_{mix} and then drops abruptly to 0 over a comparatively short window. To formalize this, suppose we have a *sequence* of Markov chains indexed by $n = 1, 2, \dots$, and write $d_n(t)$ and $t_{\text{mix}}^{(n)}(\varepsilon)$ for the corresponding distance function and mixing time of the n -th chain.

Definition 1.4 ([1, Eq. (18.3)]). The sequence of chains has a *cutoff* if, for all $\varepsilon \in (0, 1)$,

$$\lim_{n \rightarrow \infty} \frac{t_{\text{mix}}^{(n)}(\varepsilon)}{t_{\text{mix}}^{(n)}(1 - \varepsilon)} = 1. \quad (7)$$

Equivalently [1, Lemma 18.1], (7) holds if and only if $d_n(ct_{\text{mix}}^{(n)}) \rightarrow 1$ for every $c < 1$ and $d_n(ct_{\text{mix}}^{(n)}) \rightarrow 0$ for every $c > 1$. In words, when time is rescaled by the mixing time, the distance-to-stationarity profile approaches a step function as $n \rightarrow \infty$. Establishing cutoff therefore requires both an upper and a lower bound on $t_{\text{mix}}^{(n)}$ that agree to leading order—which is exactly what we will prove for the specific Markov chain we have chosen.

1.5 The top-to-random shuffle

Let S_n denote the symmetric group on $\{1, \dots, n\}$, viewed as the set of arrangements of an n -card deck. The *top-to-random shuffle* is the Markov chain on S_n whose single step is: *remove the top card and reinsert it at one of the n possible positions, each with probability $1/n$* . This is a random walk on the group S_n driven by a measure that is supported on the n “cyclic” permutations corresponding to each possible insertion position; in particular, the chain is irreducible and aperiodic, and its unique stationary distribution is the uniform measure on S_n [1, Proposition 2.12].

The intuition that this shuffle mixes slowly is correct: each step changes only the position of the top card. Nevertheless, after enough iterations every original card has been moved at least once and the deck does randomize completely. We will show that the order of magnitude of the mixing time is $n \log n$, and in fact that the chain has a cutoff at this value. The main theorem combines [1, §6.5.3] for the upper bound with [1, Proposition 7.15] for the lower bound.

Theorem 1.5. *Let $d_n(t)$ denote the function $d(t)$ from (6) in Section 1.3, applied to the top-to-random shuffle on n cards. Then:*

(i) (Upper bound) *For every $\alpha > 0$,*

$$d_n(n \log n + \alpha n) \leq e^{-\alpha}; \quad (8)$$

equivalently, for every $\varepsilon > 0$ there exists $\alpha(\varepsilon) := \log(\varepsilon^{-1})$ such that, for all $\alpha > \alpha(\varepsilon)$, $d_n(n \log n + \alpha n) \leq \varepsilon$.

(ii) (Lower bound) *For every $\varepsilon > 0$ there exists $\alpha(\varepsilon)$ such that, for all $\alpha > \alpha(\varepsilon)$ and all sufficiently large n ,*

$$d_n(n \log n - \alpha n) \geq 1 - \varepsilon. \quad (9)$$

In particular, the sequence of top-to-random shuffles has a cutoff at $n \log n$.

The remainder of the paper is devoted to the proof. In Section 2 we develop the theory of *strong stationary times*, show that an explicit random time τ_{top} is a strong stationary time for the top-to-random chain, and deduce (8) by analyzing τ_{top} as a coupon collector problem. In Section 3 we establish (9) by exhibiting an event—namely, the indicator that the bottom j cards remain in their original relative order—whose probabilities under $P^t(\text{id}, \cdot)$ and under the uniform π are far apart whenever $t \leq n \log n - \alpha n$. Section 4 combines the two.

2 The Upper Bound

The upper bound rests on the technique of *strong stationary times*: random times τ at which the chain has exactly reached its stationary distribution, and reached it *independently* of τ itself. Whenever such a time can be constructed, the tail $\mathbb{P}\{\tau > t\}$ controls the total variation distance to stationarity at time t . For the top-to-random shuffle the relevant time turns out to be distributed exactly as a coupon collector waiting time, and the upper bound follows from a standard tail estimate.

2.1 Strong stationary times

Throughout this section, $(X_t)_{t \geq 0}$ is a Markov chain on a finite state space $\mathcal{X}$ with transition matrix P and stationary distribution π , adapted to a filtration $\{\mathcal{F}_t\}$. A *stopping time* for $\{\mathcal{F}_t\}$ is a $\{0, 1, 2, \dots\}$ -valued random variable τ such that $\{\tau = t\} \in \mathcal{F}_t$ for every $t \geq 0$; allowing the filtration to be larger than the natural filtration generated by (X_t) permits stopping rules that depend on auxiliary randomness, as discussed in [1, §6.2].

Definition 2.1 ([1, §6.4]). A *strong stationary time* for the chain (X_t) started from state x is an $\{\mathcal{F}_t\}$ -stopping time τ such that, for every $t \geq 0$ and every $y \in \mathcal{X}$,

$$\mathbb{P}_x\{\tau = t, X_\tau = y\} = \mathbb{P}_x\{\tau = t\} \pi(y). \quad (10)$$

Equivalently, X_τ has distribution π and is independent of τ .

The independence requirement is what makes (10) *strong*; an ordinary stationary time only requires $X_\tau \sim \pi$. As shown in [1, Remark 6.8], summing (10) over $s \leq t$ and using the stationarity of π yields the useful identity

$$\mathbb{P}_x\{\tau \leq t, X_t = y\} = \mathbb{P}_x\{\tau \leq t\} \pi(y) \quad \text{for all } t \geq 0, y \in \mathcal{X}. \quad (11)$$

Intuitively, once τ has occurred the chain is in distribution π , and running it forward keeps it in π .

2.2 Strong stationary times bound the total variation distance

The key bound is the following, which appears as [1, Proposition 6.11].

Proposition 2.2 ([1, Proposition 6.11]). *If τ is a strong stationary time for (X_t) started at x , then for all $t \geq 0$,*

$$\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq \mathbb{P}_x\{\tau > t\}.$$

The proof in [1] proceeds via the *separation distance*, a companion to total variation distance defined by

$$s_x(t) := \max_{y \in \mathcal{X}} \left[1 - \frac{P^t(x, y)}{\pi(y)} \right]. \quad (12)$$

The bound is then split into two lemmas: the first relates the strong stationary time to $s_x(t)$, the second relates $s_x(t)$ to TV distance.

Lemma 2.3 ([1, Lemma 6.12]). *If τ is a strong stationary time for (X_t) started at x , then $s_x(t) \leq \mathbb{P}_x\{\tau > t\}$ for all $t \geq 0$.*

Proof. Fix $x \in \mathcal{X}$. For every $y \in \mathcal{X}$,

$$1 - \frac{P^t(x, y)}{\pi(y)} = 1 - \frac{\mathbb{P}_x\{X_t = y\}}{\pi(y)} \leq 1 - \frac{\mathbb{P}_x\{X_t = y, \tau \leq t\}}{\pi(y)}.$$

By the identity (11), the right-hand side equals

$$1 - \frac{\pi(y) \mathbb{P}_x\{\tau \leq t\}}{\pi(y)} = \mathbb{P}_x\{\tau > t\}.$$

Taking the maximum over $y \in \mathcal{X}$ gives the claim. $\square$

Lemma 2.4 ([1, Lemma 6.16]). *For all $t \geq 0$ and $x \in \mathcal{X}$, $\|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq s_x(t)$.*

Proof. Applying the equivalent form (3) of total variation distance with $\mu = \pi$ and $\nu = P^t(x, \cdot)$,

$$\begin{aligned} \|P^t(x, \cdot) - \pi\|_{\text{TV}} &= \sum_{y: \pi(y) \geq P^t(x, y)} [\pi(y) - P^t(x, y)] \\ &= \sum_{y: \pi(y) \geq P^t(x, y)} \pi(y) \left[1 - \frac{P^t(x, y)}{\pi(y)}\right] \\ &\leq \max_{y \in \mathcal{X}} \left[1 - \frac{P^t(x, y)}{\pi(y)}\right] = s_x(t), \end{aligned}$$

where the inequality uses $\sum_y \pi(y) \leq 1$. $\square$

Proof of Proposition 2.2. Combine Lemmas 2.3 and 2.4. $\square$

2.3 The time τ_{top} is a strong stationary time

We now return to the top-to-random shuffle. Label the cards so that the deck initially reads $(c_1, c_2, \dots, c_n)$ from bottom to top, and let $X_t \in S_n$ denote the arrangement at time t , with $X_0 = \text{id}$. Define

$$\tau_{\text{top}} := 1 + \inf\{t \geq 0 : c_1 \text{ is at the top of } X_t\}. \quad (13)$$

Equivalently, τ_{top} is the first time at which c_1 is moved by a shuffle.

Proposition 2.5 ([1, Proposition 6.1]). *At any time t , given that there are k cards beneath c_1 , the $k!$ possible orderings of those cards are equally likely. Consequently τ_{top} is a strong stationary time for the top-to-random shuffle started at id .*

Proof. We argue by induction on t . At $t = 0$ there are no cards beneath c_1 , and the claim is trivially valid. Suppose the claim holds at time t , and that there are k cards beneath c_1 . There are two possibilities at time $t + 1$: either a card is placed under c_1 , or not. In the second case the cards under c_1 remain in random order. In the first case, given that a card is placed under c_1 , each of the $k + 1$ possible locations for the card is equally likely, and so each of the $(k + 1)!$ orderings are equiprobable.

The conclusion follows from the inductive claim applied at time τ_{top} : at that moment c_1 has just been moved from the top into a uniformly chosen one of n positions, while the other $n - 1$ cards are in uniformly random order beneath; thus $X_{\tau_{\text{top}}}$ is uniform on S_n , conditional on $\tau_{\text{top}} = t$ for every t . This is precisely (10). $\square$

2.4 Coupon collector and the upper bound

It remains to estimate $\mathbb{P}\{\tau_{\text{top}} > t\}$. The key observation is that τ_{top} has the same distribution as a coupon collector waiting time. In the standard coupon collector problem [1, §2.2], a collector draws coupons uniformly at random with replacement from a set of n types, and the relevant random variable τ is the first time at which all n types have been seen.

To see the connection, track the position of c_1 under the top-to-random shuffle. Whenever c_1 has k cards beneath it ($0 \leq k \leq n-1$), the next shuffle places the current top card into a uniformly chosen one of the n positions; exactly $k+1$ of those positions lie at or below c_1 , so c_1 rises by one with probability $(k+1)/n$. The waiting time for c_1 to rise from having k cards beneath to having $k+1$ is therefore geometric with parameter $(k+1)/n$, and these waiting times are independent. Hence

$$\tau_{\text{top}} = G_1 + G_2 + \cdots + G_n, \quad G_k \sim \text{Geom}(k/n) \text{ independent.} \quad (14)$$

This decomposition matches exactly the standard decomposition of the coupon collector time given in [1, Eq. (2.5)], so $\tau_{\text{top}} \stackrel{d}{=} \tau$ for τ a coupon collector on n types. In particular, since $\mathbb{E}[\text{Geom}(k/n)] = n/k$,

$$\mathbb{E}[\tau_{\text{top}}] = \sum_{k=1}^n \frac{n}{k} = nH_n = n \log n + O(n), \quad (15)$$

where H_n is the n -th harmonic number. This already pinpoints $n \log n$ as the relevant order of magnitude; a Chebyshev bound applied to (14) would yield a tail estimate at this scale, but a strictly sharper one is available by direct union bound.

Proposition 2.6 ([1, Proposition 2.4]). *Let τ be a coupon collector random variable on n types. For every $c > 0$,*

$$\mathbb{P}\{\tau > \lceil n \log n + cn \rceil\} \leq e^{-c}.$$

Proof. Let A_i be the event that the i -th type does not appear among the first $\lceil n \log n + cn \rceil$ draws. Then

$$\mathbb{P}\{\tau > \lceil n \log n + cn \rceil\} = \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

Since each draw misses type i independently with probability $1 - 1/n$,

$$\sum_{i=1}^n \mathbb{P}(A_i) = n \left(1 - \frac{1}{n}\right)^{\lceil n \log n + cn \rceil} \leq n \exp\left(-\frac{n \log n + cn}{n}\right) = e^{-c}. \quad \square$$

Corollary 2.7. *For the top-to-random shuffle on n cards and every $\alpha > 0$,*

$$d_n(n \log n + \alpha n) \leq e^{-\alpha}.$$

Proof. By Propositions 2.2 and 2.5, $d_n(t) \leq \mathbb{P}\{\tau_{\text{top}} > t\}$. Setting $t = \lceil n \log n + \alpha n \rceil$ and applying Proposition 2.6 to the coupon-collector-distributed τ_{top} gives the bound; since d_n is non-increasing in t , we may replace the ceiling by $n \log n + \alpha n$. $\square$

3 The Lower Bound

The upper bound shows that after about $n \log n$ shuffles the deck is close to uniform. To establish cutoff we need to show that significantly *fewer* shuffles do *not* suffice. The argument, due to

Aldous and Diaconis and reproduced in [1, Proposition 7.15], uses only the definition of total variation distance: we exhibit a single event $A \subseteq S_n$ on which the time- t distribution $P^t(\text{id}, \cdot)$ and the uniform distribution π disagree substantially. By definition of TV distance,

$$\|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \geq |P^t(\text{id}, A) - \pi(A)|. \quad (16)$$

3.1 The bottom-cards event

Following [1, Eq. (7.24)], for an integer j with $2 \leq j \leq n$ define

$$A_j := \{ \sigma \in S_n : \text{the original bottom } j \text{ cards are in their original relative order in } \sigma \}. \quad (17)$$

That is, A_j holds when $c_1, c_2, \dots, c_j$ still appear from bottom to top in that order—they may have moved up the deck and other cards may be interleaved among them, but the lowest of the j remains lower than the next, and so on. Under the uniform distribution π on S_n , all $j!$ relative orders of these j cards are equally likely, so

$$\pi(A_j) = \frac{1}{j!}. \quad (18)$$

To bound $P^t(\text{id}, A_j)$ from below, let τ_j be the time required for the card initially j -th from the bottom (namely c_j) to reach the top. As long as $t \leq \tau_j$, none of $c_1, \dots, c_j$ has been touched by a shuffle—each shuffle moves the current top card, and none of $c_1, \dots, c_j$ is at the top before time τ_j —so they remain in their original relative order. Hence $X_t \in A_j$ on $\{\tau_j \geq t\}$, giving

$$P^t(\text{id}, A_j) \geq \mathbb{P}\{\tau_j \geq t\}. \quad (19)$$

3.2 Mean and variance of τ_j

The card c_j starts at position j from the bottom. When it is at position i from the bottom (for $i < n$), the next shuffle moves c_j up by one if and only if the current top card is reinserted into one of the i slots at or below c_j , an event of probability i/n . Therefore

$$\tau_j = \sum_{i=j}^{n-1} \tau_{j,i},$$

where $\tau_{j,i}$ is the time it takes the card initially j -th from the bottom to ascend from position i to position $i+1$. The variables $\{\tau_{j,i}\}_{i=j}^{n-1}$ are independent, and $\tau_{j,i}$ is geometrically distributed with parameter $p = i/n$, so $\mathbb{E}[\tau_{j,i}] = n/i$ and $\text{Var}(\tau_{j,i}) < n^2/i^2$. Following [1, Eqs. (7.25)–(7.26)],

$$\mathbb{E}[\tau_j] = \sum_{i=j}^{n-1} \frac{n}{i} \geq n \int_j^n \frac{dx}{x} = n(\log n - \log j), \quad (20)$$

and using the telescoping identity $\sum_{i=j}^{\infty} 1/[i(i-1)] = 1/(j-1)$ (valid for $j \geq 2$),

$$\text{Var}(\tau_j) \leq n^2 \sum_{i=j}^{\infty} \frac{1}{i(i-1)} = \frac{n^2}{j-1}. \quad (21)$$

3.3 Proof of the lower bound

Proposition 3.1 ([1, Proposition 7.15]). *For every $\varepsilon > 0$ there exists $\alpha(\varepsilon)$ such that $\alpha > \alpha(\varepsilon)$ implies*

$$\liminf_{n \rightarrow \infty} d_n(n \log n - \alpha n) \geq 1 - \varepsilon.$$

Proof. Set $t_n(\alpha) := n \log n - \alpha n$. Using the bounds (20) and (21), together with Chebyshev's inequality, for $\alpha \geq \log j + 1$,

$$\mathbb{P}\{\tau_j < t_n(\alpha)\} \leq \mathbb{P}\{\tau_j - \mathbb{E}[\tau_j] < -n(\alpha - \log j)\} \leq \frac{\text{Var}(\tau_j)}{n^2(\alpha - \log j)^2} \leq \frac{1}{j-1}.$$

If $\tau_j \geq t_n(\alpha)$, then by (19) the original j bottom cards remain in their original relative order at time $t_n(\alpha)$, so

$$P^{t_n(\alpha)}(\text{id}, A_j) \geq \mathbb{P}\{\tau_j \geq t_n(\alpha)\} \geq 1 - \frac{1}{j-1} \quad \text{for } \alpha \geq \log j + 1.$$

On the other hand, $\pi(A_j) = 1/j! \leq 1/(j-1)$, so by (16), for $\alpha \geq \log j + 1$,

$$d_n(t_n(\alpha)) \geq \|P^{t_n(\alpha)}(\text{id}, \cdot) - \pi\|_{\text{TV}} \geq P^{t_n(\alpha)}(\text{id}, A_j) - \pi(A_j) > 1 - \frac{2}{j-1}. \quad (22)$$

Take $j = \lceil e^{\alpha-1} \rceil$. Provided $n \geq e^{\alpha-1}$ (so $j \leq n$), we have

$$d_n(t_n(\alpha)) > g(\alpha) := 1 - \frac{2}{\lceil e^{\alpha-1} \rceil - 1},$$

hence $\liminf_{n \rightarrow \infty} d_n(t_n(\alpha)) \geq g(\alpha)$. Since $g(\alpha) \rightarrow 1$ as $\alpha \rightarrow \infty$, given $\varepsilon > 0$ we may choose $\alpha(\varepsilon)$ such that $g(\alpha) \geq 1 - \varepsilon$ for all $\alpha > \alpha(\varepsilon)$. $\square$

4 The Cutoff Phenomenon

Combining Corollary 2.7 and Proposition 3.1 proves Theorem 1.5. We extract the cutoff statement explicitly.

Corollary 4.1. *The sequence of top-to-random shuffles exhibits a cutoff at $n \log n$.*

Proof. Fix $\varepsilon \in (0, 1)$. By Corollary 2.7, $d_n(n \log n + \alpha n) \leq e^{-\alpha} < \varepsilon$ once $\alpha > \log(\varepsilon^{-1})$, so $t_{\text{mix}}^{(n)}(\varepsilon) \leq n \log n + \log(\varepsilon^{-1})n$ for every n . By Proposition 3.1, for $\alpha > \alpha(\varepsilon)$ we have $\liminf_n d_n(n \log n - \alpha n) \geq 1 - \varepsilon$, so $t_{\text{mix}}^{(n)}(1 - \varepsilon) \geq n \log n - \alpha n$ for all sufficiently large n . Combining,

$$\frac{n \log n - \alpha n}{n \log n + \log(\varepsilon^{-1})n} \leq \frac{t_{\text{mix}}^{(n)}(1 - \varepsilon)}{t_{\text{mix}}^{(n)}(\varepsilon)} \leq \frac{n \log n + \log(\varepsilon^{-1})n}{n \log n - \alpha n},$$

and both extremes converge to 1 as $n \rightarrow \infty$. By Definition 1.4, the sequence has a cutoff. $\square$

The qualitative picture is the one suggested by Figure 18.1 of [1]: when time is rescaled by $n \log n$, the function d_n approaches a step function that is identically 1 on $[0, 1)$ and identically 0 on $(1, \infty)$.

The technique generalizes. Strong stationary times provide upper bounds for many random walks on groups [1, Chapter 6]; lower bounds following the pattern above (or its variance-based generalization, [1, Proposition 7.9]) apply to many of the same examples [1, Chapter 7]; and the cutoff phenomenon, first identified through examples like the one analyzed here, has since been verified for a wide variety of chains and remains an active area of research [1, Chapter 18].

References

- [1] David A. Levin and Yuval Peres, with contributions by Elizabeth L. Wilmer. *Markov Chains and Mixing Times*, second edition. American Mathematical Society, Providence, RI, 2017.